October 12, 2026

Dr. Jenahvive Morgan
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Dear Dr. Morgan,

Attached is a copy of Section 2, Team 7's Project #1 Final Report for EGR 100.

This report describes the design and prototyping process for a localized mobile application, "Night Owl Dining," engineered to resolve late-night dining access challenges faced by students at Michigan State University. The multi-screen graphic user interface was optimized for real-time navigation by integrating user-preference selection algorithms. Programmatic constraints and execution environments were systematically evaluated and analyzed through localized participant beta testing.

Utilizing the MIT App Inventor development platform, the final architecture was engineered as a modular three-screen system relying on a unified localized database and an asset registry. Screen 1 establishes a centralized user dashboard featuring dynamic sorting configurations for sit-down and take-out dining integrated with native clock functionality to automatically filter open locations by day and hour, enabling users to execute custom selection queries and launch external merchant web interfaces. Screen 2 integrates a real-time user-directed Web Viewer interface paired with custom logic blocks to initialize routing requests. Screen 3 deploys a dedicated Web Viewer component running a Google Maps API framework, which dynamically reads user and restaurant coordinates to compute real-time driving durations. When evaluated against design criteria, the prototype successfully satisfied all structural economic parameters and technical data-maintenance limitations.

I hope this report meets your requirements, and any questions will gladly be addressed by email.

Sincerely,

Andrew Sreseli

David Parker

Anthony Marchese

Daniel Wang

INTRODUCTION

The purpose of this project is to solve a localized, real-time navigation problem faced by Michigan State University students: locating open food options after major campus dining halls close and tracking variable late-night restaurant operating hours along Grand River Avenue. Students studying late at the library, working on engineering projects in computer labs, or returning from late-night student activities currently lack a centralized, real-time utility.

While research shows time, price, and location are critical for dining recommendations [1], popular commercial platforms frequently fail to capture spatial efficiency and access to essential resources in dense college environments. Additionally, mobile location data indicates that travel willingness heavily shapes dining choices [2]. The *Night Owl Dining* mobile utility addresses critical operational inefficiencies by providing an accessible, real-time optimization tool engineered to bridge this disparity.

METHODS

The *Night Owl Dining* application was developed and prototyped using the MIT App Inventor programming environment [3]. The software architecture was structured as a modular three-screen system utilizing a unified asset registry.
Screen 1 serves as the main user dashboard, featuring categorized restaurant lists for sit-down (Figure 1) and take-out (Figure 2) dining options. Users interact with selection buttons to choose a specific venue, which triggers logic blocks to launch external merchant links.

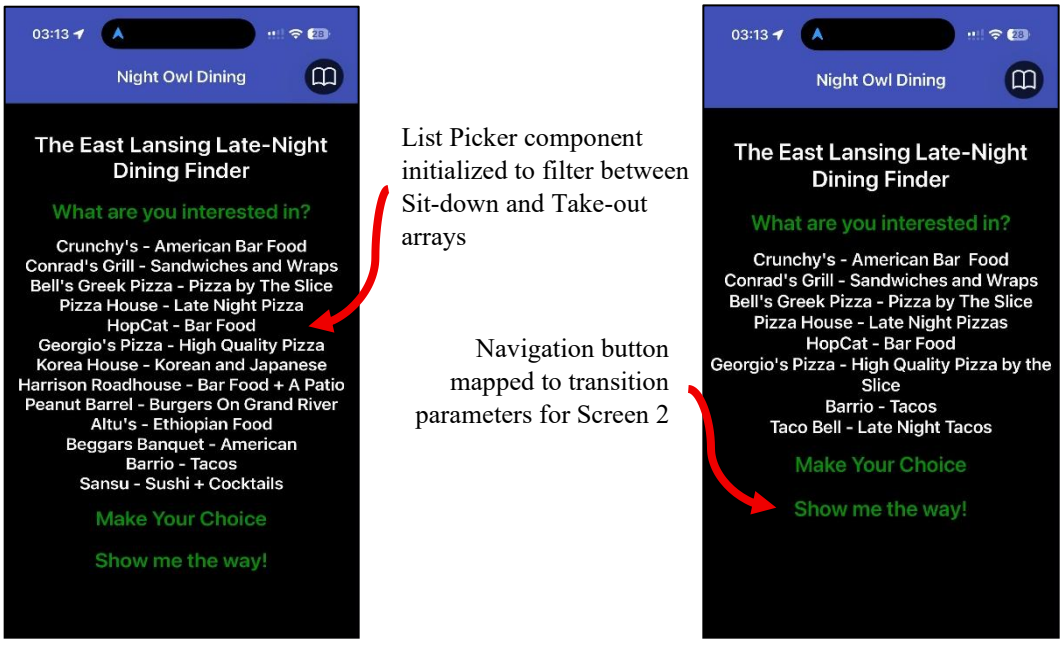


Figure 1: Screen 1 user interface layout showcasing localized sit-down dining options.

Figure 2: Screen 1 user interface layout showcasing localized take-out dining options.

Screen 2 (Figure 3) integrates a real-time Web Viewer component paired with a dedicated button to initialize direction queries [3]. Screen 3 (Figure 4) embeds a secondary Web Viewer interface running a live Google Maps webpage structure. This component reads the user's manual selection inputs to display directions and calculate driving times to the chosen restaurant location along Grand River Avenue. The layout of the graphic user interface and its underlying block logic configuration are shown in designated figures.

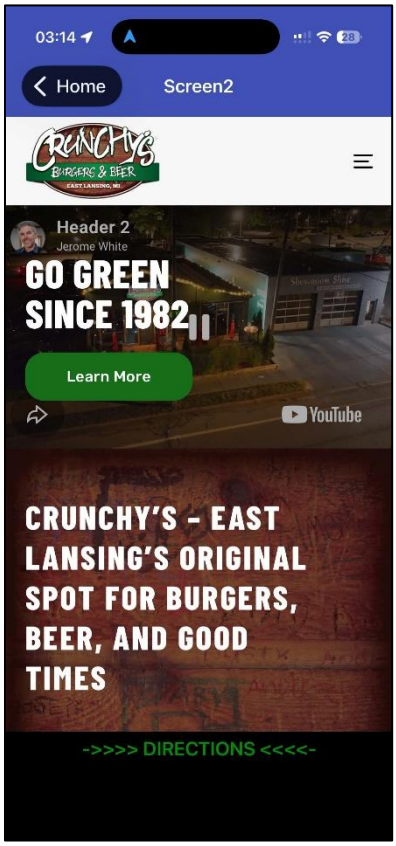


Figure 3: Screen 2 layout initializing the dynamic Web Viewer merchant interface.

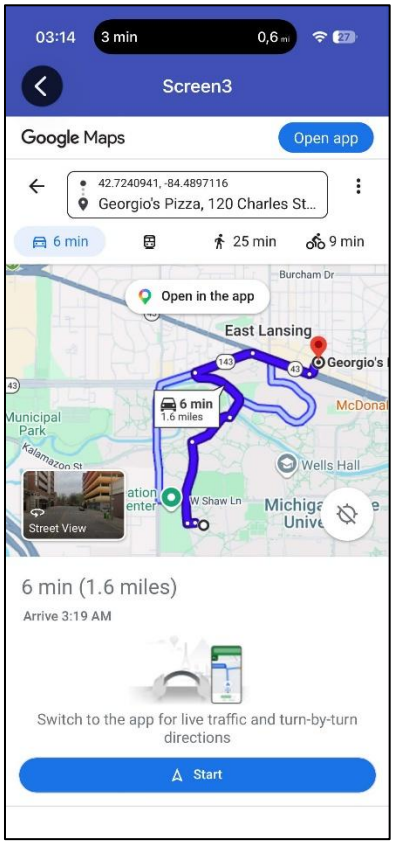


Figure 4: Screen 3 layout displaying routing via the Google Maps component.

The backend logic for Screen 1 (Figure 5) executed a conditional sorting routine based on user inputs and real-time temporal parameters. The system parsed the initial selection from the primary List Picker component and cross-referenced a universally integrated Clock component to filter options dynamically by both the current day of the week and the specific hour of operation. The system then populated a secondary array containing only the active, open merchant assets that correlated with both the user's specified dining preference and the current time constraints. Upon interaction with the secondary List Picker, the student's final restaurant choice was systematically committed to local storage utilizing a persistent Tiny DB component. Execution of the navigation command button subsequently initialized a screen transition to Screen 2.

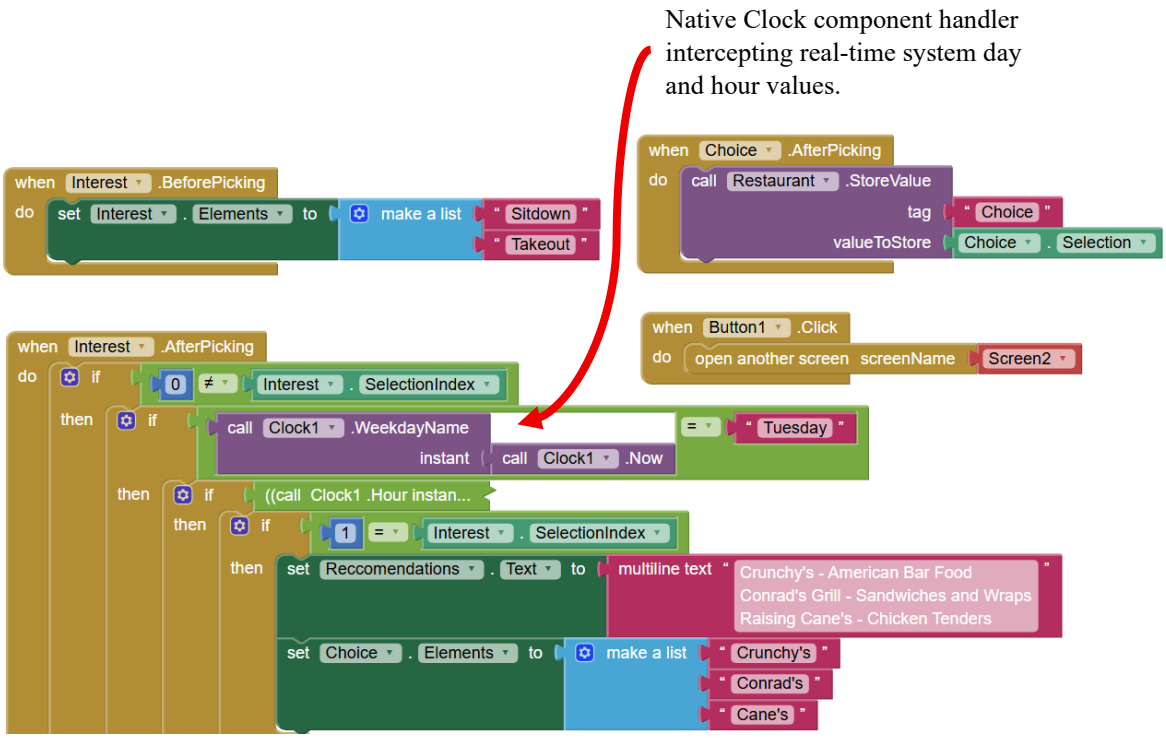


Figure 5: Event-driven block logic operating the Screen 1 selection and data-storage routines.

The control logic for Screen 2 (Figure 6) retrieved the stored string token from the Tiny DB registry to identify the user's target restaurant. The system then invoked a Set Web Viewer action block to direct the embedded Web Viewer interface to load the verified website of the selected merchant. A secondary command button was integrated to initialize a screen shift to Screen 3 when prompted by the user.

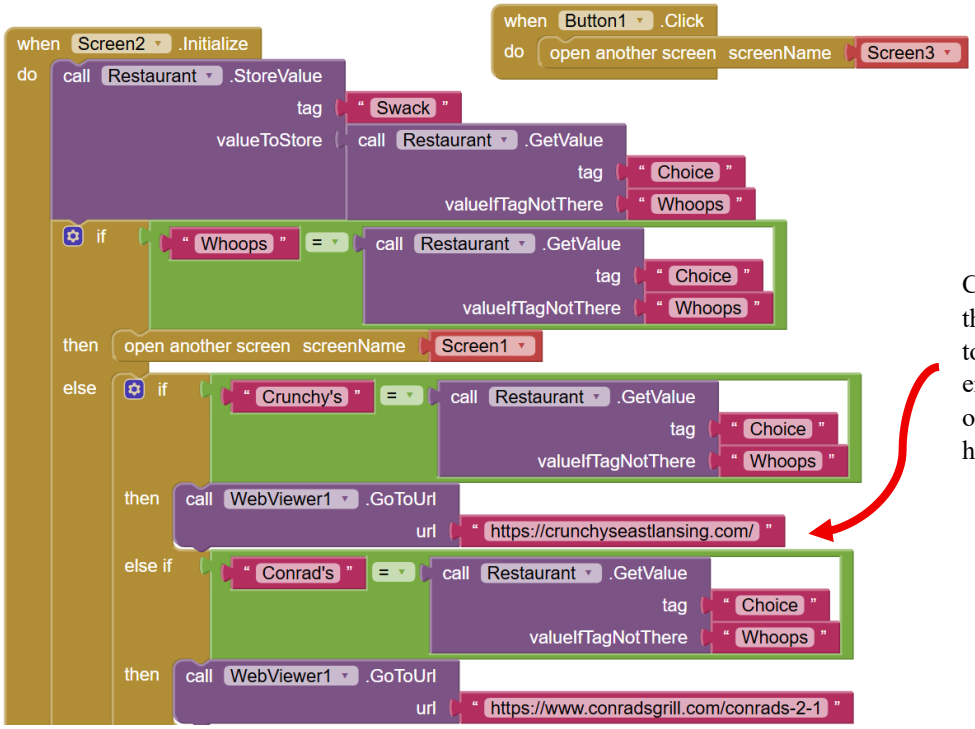


Figure 6: System logic block diagram operating the Screen 2 Web Viewer interface configurations.

The execution for Screen 3 (Figure 7) queried the Tiny DB component a final time to verify the persistent restaurant identity string. Following verification, the system compiled a custom URL string to direct the secondary Web Viewer component to open a live Google Maps framework displaying explicit navigation parameters to the targeted Grand River Avenue location.

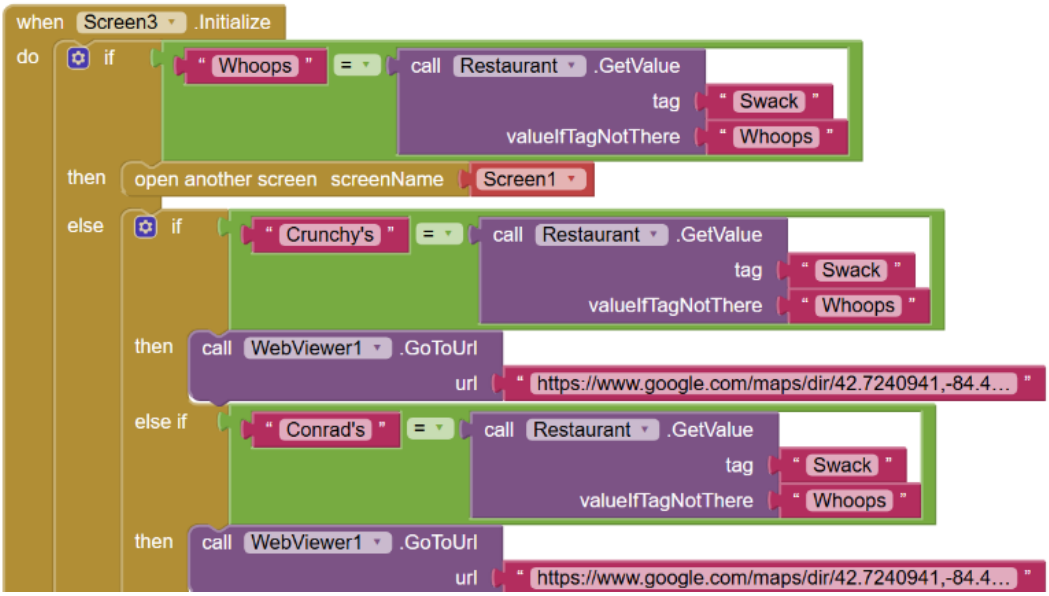


Figure 7: Final routing logic blocks executing the Screen 3 Google Maps API interface integration.

Alternative software architectures were evaluated during development. An initial iteration relied on a manual list layout with static text blocks, which was rejected because it required students to manually verify operating hours. Based on testing and user evaluation feedback, a native software Clock component was integrated to automatically capture real-time temporal parameters. This adjustment allowed the block logic to filter out closed venues dynamically by cross-referencing active operating hours, improving the overall functionality of the application.

RESULTS

Surveys and usability testing are established methods for evaluating users ’experiences with interactive systems [5]. To evaluate user requirements and validate application performance, a targeted beta testing pool consisting of 30 Michigan State University students was systematically surveyed. Quantitative user feedback metrics were compiled into a centralized asset database to analyze baseline late-night demand along Grand River Avenue alongside user satisfaction trends.

To satisfy explicit engineering documentation standards, two distinct graphical metrics were compiled. First, to display discrete responses relative to individual participants, Figure 8 illustrates the individual App Helpfulness Rating plotted cleanly across each unique respondent node (1 to 30) along the linear horizontal scale. The spatial layout highlights that despite expected individual variance across the cohort, helpfulness evaluations remain heavily concentrated at high functional tiers, achieving a parameter mean of $\mu = 3.83$ paired with a tight standard deviation of $\sigma = 0.82$.

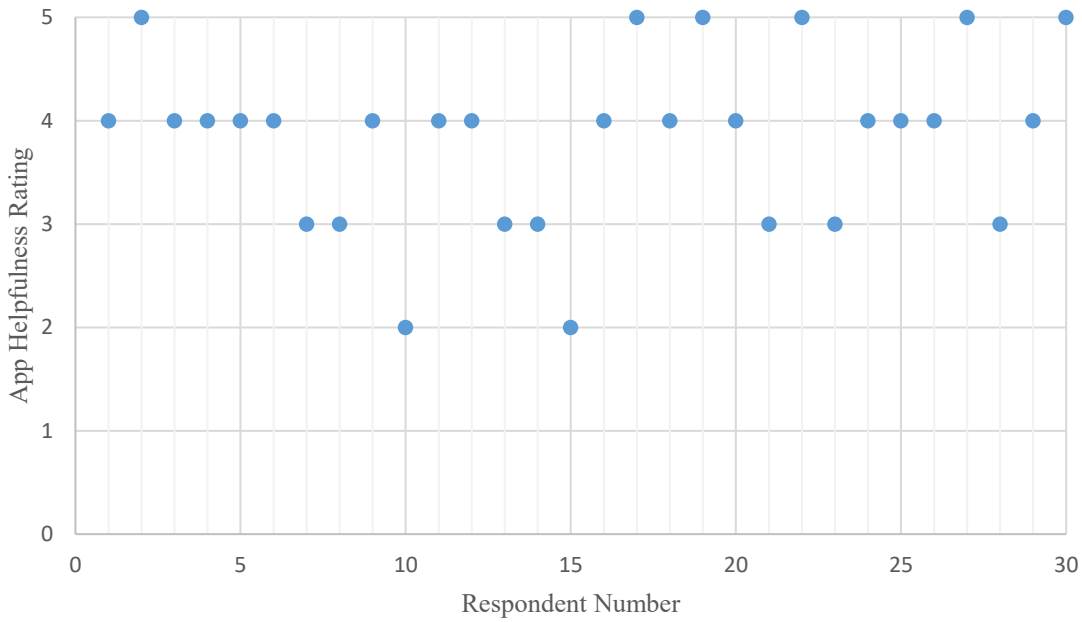


Figure 8: Individual App Helpfulness Rating Distributed by Participant Trial Number

Individual respondent entries were mapped onto a coordinate matrix to evaluate student demands against performance metrics (Figure 9). To ensure all data points remain distinct and legible within strict map borders, an inward geometric offset was applied to overlapping integer values. The spatial distribution demonstrates strong positive clustering between high problem frequency metrics and elevated software satisfaction. Notably, heavy-user student demographics registering a maximum operational demand factor of 5.00 consistently recorded application satisfaction scores at or above 4.00, validating that the multi-screen architecture scales effectively along Grand River Avenue.

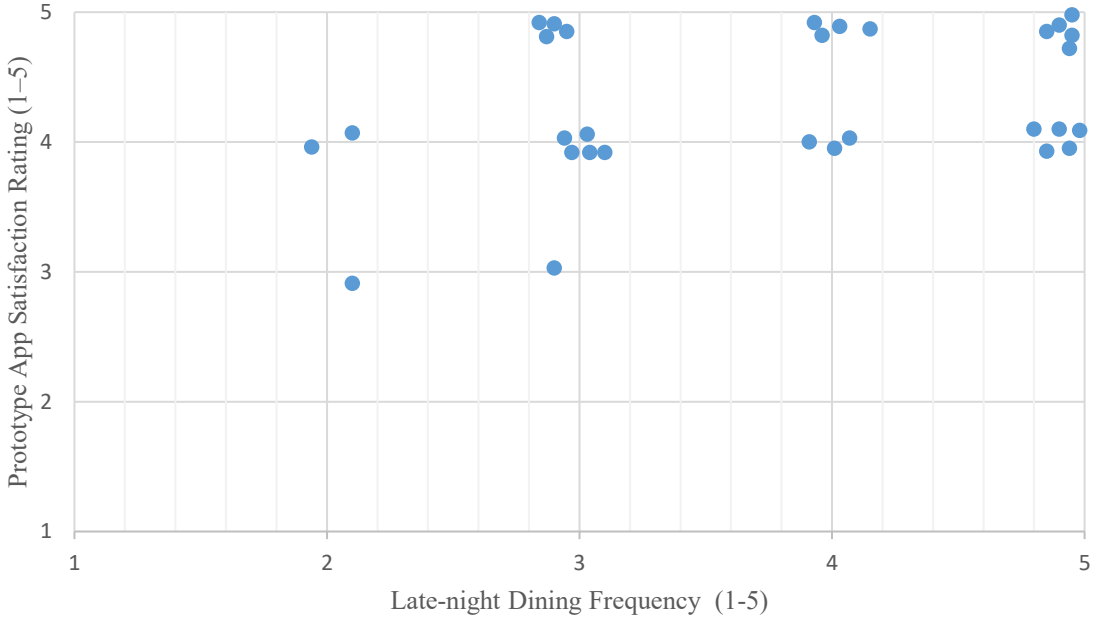


Figure 9: Student App Satisfaction Rating as a Function of Late-Night Dining Problem Frequency

Individual user response data entries are comprehensive and itemized below in Table 1.

Table 1: Table consisting of all answers of responders who participated in the survey

Respondent Number	Late-night Dining Frequency (1-5)	App Helpfulness Rating (1-5)	App Satisfaction Rating (1-5)
1	5	4	5
2	5	5	5
3	4	4	4
4	3	4	4
5	3	4	4
6	5	4	5
7	5	3	4
8	3	3	4
9	5	4	4
10	2	2	4
11	5	4	4
12	4	4	5
13	5	3	4
14	3	3	5
15	3	2	3
16	3	4	5
17	4	5	5
18	4	4	4
19	5	5	5
20	3	4	5
21	3	3	4
22	4	5	5
23	2	3	3
24	3	4	5
25	5	4	5
26	3	4	4
27	4	5	5
28	2	3	4
29	5	4	4
30	4	5	4

A summary of the compiled statistical parameters are detailed in Table 2.

Table 2: Statistical Survey Parameters for Target Student Focus Group

Survey Parameter	Sample Size (N)	Parameter mean (μ)	Standard Deviation (σ)
Late-night Dining Frequency (1-5)	30	3.80	1.01
App Helpfulness Rating (1-5)	30	3.83	0.82
App Satisfaction Rating (1-5)	30	4.37	0.60

As detailed in Table 1, the targeted user population reported a high frequency of late-night dining demand ($\mu = 3.80$) with a standard deviation of 1.01, confirming a widespread and frequent need for a centralized campus information utility. Following interaction trials with the software prototype, the application achieved an engineering validation metric characterized by an App Helpfulness Rating mean of 3.83 out of 5.00 ($\sigma = 0.82$).
Furthermore, the overall user software interface and multi-screen system workflow yielded a very high App Satisfaction Rating ($\mu = 4.37$). The low standard deviation recorded within this parameter ($\sigma = 0.60$) demonstrates consistency in user utility, confirming that the modular asset architecture successfully satisfied data accessibility requirements under standard campus operational constraints.

CONCLUSION

The objective of this project was to engineer a localized software utility, *Night Owl Dining*, to resolve the real-time spatial navigation and informational disparities experienced by students seeking operational food resources after standard campus dining hall closures at Michigan State University. By utilizing the MIT App Inventor programming platform, a modular three-screen user interface was successfully prototyped and validated against technical and student demographic constraints.

Evaluation of the final system design using survey testing metrics from a target student focus group demonstrated high operational success. The software achieved a mean App Helpfulness Rating of 3.83 out of 5.00 and an App Satisfaction Rating of 4.37 out of 5.00, proving that the localized menu filtering systems and embedded Web Viewer map routing mechanics functioned with high consistency across the testing pool. While the evaluation was structurally limited by an academic project sample constraint of 30 participants, the strong positive correlation between high user problem frequencies and elevated satisfaction metrics confirms that the application architecture successfully resolved information gaps along Grand River Avenue. Future iterations will expand backend database structures to integrate automated, real-time operating hour update blocks.

REFERENCES

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[4] J. Lazar, J. H. Feng, and H. Hochheiser, *Research Methods in Human-Computer Interaction*, 2nd ed. Cambridge, MA, USA: Morgan Kaufmann, 2017.